

Statistical Analysis Supporting the Assessment of Human–Elephant Coexistence

S1. Data Preparation and Variable Coding

S1.1 Survey Dataset

The statistical analyses were based on household-level socio-economic survey data collected from 744 respondents across 140 villages in the Bandhavgarh–Sanjay landscape during January 2024–June 2025. Survey responses were entered into a structured database and subsequently imported into IBM SPSS Statistics for statistical analysis. Variables relevant to community willingness to coexist with elephants were coded according to their measurement scale and analytical requirements. Categorical predictors were represented using indicator/dummy coding for binary logistic regression, with appropriate reference categories specified for interpretation of regression coefficients.

S1.2 Dependent Variable

Table S1. Dependent variable used in the statistical analyses

Variable	SPSS variable name	Role	Coding
Willingness to coexist	Coexistence	Dependent variable	0 = No; 1 = Yes

S1.3 Independent Variables

Table S2. Independent variables included in the statistical analyses

S. No.	Variable	SPSS variable name	Role	Measurement
1	Village location	Village_location	Predictor	Categorical
2	Age	Age	Predictor	Categorical
3	Gender	Gender	Predictor	Categorical
4	Tribal status	Tribal_nonTribal	Predictor	Binary categorical
5	Education	Education	Predictor	Categorical
6	Occupation	Occupation	Predictor	Categorical
7	Income	Income	Predictor	Categorical
8	NTFP dependence	Dependents	Predictor	Categorical
9	Distance from forest	Distance_from_forest	Predictor	Categorical
10	Elephant group type	Herd_tusker	Predictor	Categorical
11	Understanding of elephant ecology	Understanding_level	Predictor	Categorical
12	HEC frequency	HEC_Frequency	Predictor	Categorical

S1.4 Variable Coding for Logistic Regression

Table S3. Coding and reference categories of predictors used in binary logistic regression

Predictor	Category	Numerical code	Reference category
Village location	Corridor Village	0	Protected Area*
	STR Village	1	

	BTR Village	2	
Age	Young	0	Young*
	Middle-aged	1	
	Elderly	2	
Gender	Male	0	Male*
	Female	1	
Tribal status	Tribal	0	Non-tribal*
	Non-tribal	1	
Education	Illiterate	0	Illiterate*
	Primary	1	
	Secondary and Above	2	
Occupation	Farmer+farm-Labourers	0	Agriculture*
	Farmer+Job/Business	1	
	Labour	2	
Income	Low income	0	< 5000
	Medium income	1	5000 - 10000
	High income	2	>10000
NTFP dependence	Yes	0	No*
	No	1	
Distance from forest	<500 m	0	>500 m*
	>500 m	1	
Elephant group type	Herd	0	Herd*
	Solitary tusker	1	
	Both	2	
Understanding level	Low	0	Low*
	Moderate	1	
	High	2	
HEC frequency	Low	0	Low Conflict zone
	Moderate	1	Moderate Conflict zone
	High	2	High Conflict zone

S1.5 Data Screening and Preparation

Prior to statistical analysis, survey records were screened for completeness and consistency. Categorical responses were coded numerically and organised according to the requirements of the statistical tests. Variables intended for binary logistic regression were converted into indicator variables where required, and reference categories were specified to facilitate interpretation of regression coefficients. The final analytical dataset was then used for Pearson's chi-square tests and binary logistic regression.

S2. Pearson's Chi-square Analysis

Pearson's chi-square test of association was performed to examine the relationships between categorical socio-demographic characteristics, human–elephant conflict (HEC) variables, and coexistence-related responses among surveyed households. The analysis was used to determine whether the distribution of responses differed significantly across categories of the explanatory variables. The test results provided an initial assessment of associations between household

characteristics, conflict exposure, elephant-related experiences, and willingness to coexist before subsequent binary logistic regression analysis.

S2.2 Input Data and Variables

The chi-square analysis was conducted using the household-level survey dataset comprising 744 respondents from 140 villages. The analysis included categorical variables describing village location, age, gender, tribal status, education, occupation, income, NTFP dependence, distance from the forest boundary, elephant group type, understanding of elephant ecology, HEC frequency, and coexistence-related responses. The dependent coexistence response was coded as a categorical variable representing willingness to coexist with elephants. The same survey variables were subsequently considered in the binary logistic regression analysis.

S2.3 Statistical Procedure

Pearson's chi-square (χ^2) test of association was used to assess the independence between categorical variables. For each comparison, observed frequencies were compared with the frequencies expected under the null hypothesis of no association. Statistical significance was assessed at $\alpha = 0.05$. The chi-square statistic was calculated as:

where O represents the observed frequency and E represents the expected frequency. Contingency tables were examined to assess compliance with the expected-frequency assumption. Where required, categories were combined to obtain adequate expected cell frequencies before conducting the test. The resulting χ^2 statistic, degrees of freedom (df), sample size (N), and p-value were recorded for each variable association.

S2.4 Complete Pearson's Chi-square Test Output

The complete results of the Pearson's chi-square tests are presented in Table S4. The table reports the sample size (N), Pearson's chi-square statistic (χ^2), degrees of freedom (df), and corresponding p-value for each tested association. Associations with $p < 0.05$ were considered statistically significant. The supplementary table provides the complete statistical output underlying the associations reported in the main manuscript.

TABLE 1: RESULTS OF PEARSON'S CHI-SQUARE TESTS SHOWING ASSOCIATIONS BETWEEN SOCIO-DEMOGRAPHIC VARIABLES, HUMAN–ELEPHANT CONFLICT CHARACTERISTICS, MITIGATION MEASURES, AND COEXISTENCE-RELATED RESPONSES.

S.No.	Independent Variable	Associated Variable	N	χ^2	df	p-value	Statistical Significance
1.	HEC Frequency	Understanding Towards Elephants	744	57.075	6	0.000	Significant
		Willingness to Coexist	744	85.618	4	0.001	Significant
		Willingness to Relocate	744	8.735	2	0.031	Significant
		Willingness to change crop pattern	742	10.362	2	0.006	Significant
		Property Damage	744	2.498	2	0.000	Significant
		Crop loss %	530	31.861	2	0.000	Significant
2.	Elephant Group Type	Crop loss%	530	4.780	2	0.092	Not Significant
		Property Damage	690	33.634	2	0.000	Significant
		Stage of Crop	515	66.609	6	0.000	Significant

		Season	688	169.914	6	0.000	Significant
		Time	674	13.969	4	0.007	Significant
3.	Season	Crop Loss %	530	15.388	3	0.002	Significant
		Property Damage	691	34.655	3	0.000	Significant
4.	Level of Understanding Towards Elephants	Age	744	16.095	6	0.013	Significant
		Education	744	35.098	6	0.000	Significant
		Property Damage	744	11.976	3	0.007	Significant
		Crop Loss %	530	2.222	3	0.528	Not Significant
5.	Education	Willingness to Coexist	744	67.038	4	0.000	Significant
6.	Mitigations taken	Effectiveness	504	39.171	2	0.000	Significant

S3. Multicollinearity Assessment Using Linear Regression

S3.1 Objective

Multicollinearity among the explanatory variables considered for the subsequent binary logistic regression analysis was assessed prior to model fitting. The purpose of this assessment was to identify highly correlated predictors that could inflate the variance of regression coefficients and affect the stability and interpretation of the logistic regression model.

S3.2 Input Variables

A preliminary linear regression analysis was performed using the household-level coexistence response (Coexistence) as the dependent variable and the candidate socio-demographic, spatial, conflict-related, and elephant-related predictors as explanatory variables. The predictors included village location, age, tribal status, education, occupation, income, NTFP dependence, distance from the forest, elephant group type, understanding level, and HEC frequency.

S3.3 Statistical Procedure

Multicollinearity was evaluated using the Tolerance and Variance Inflation Factor (VIF) statistics obtained from the coefficient output of the preliminary linear regression model. Tolerance represents the proportion of variance in a predictor that is not explained by the remaining predictors, whereas VIF quantifies the degree to which the variance of a regression coefficient is inflated because of correlations among explanatory variables. VIF values below 5 were considered indicative of the absence of substantial multicollinearity among the candidate predictors. The resulting diagnostics were examined before proceeding to binary logistic regression.

S3.4 VIF Output

Table S4. Multicollinearity diagnostics obtained from the preliminary linear regression analysis

Coefficients ^a							
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Collinearity Statistics
		B	Std. Error	Beta			Tolerance
1	(Constant)	.531	.059		8.964	.000	

PA_NonPA_Village	-.327	.041	-.322	- 7.897	.000	.678	1.474
Age_group_class	.019	.022	.031	.869	.385	.898	1.114
Tribal_NonTribal	.112	.040	.095	2.783	.006	.967	1.035
Education_Class	-.046	.024	-.073	- 1.885	.060	.761	1.314
Occupation_Class	.014	.029	.017	.482	.630	.879	1.137
Income_Class	-.022	.031	-.025	-.711	.477	.889	1.125
NTFP_Dependens	-.019	.071	-.010	-.268	.788	.858	1.166
Village_distance_From_Forest	.126	.045	.100	2.821	.005	.904	1.106
Elephant_Group_type	-.106	.026	-.147	- 4.133	.000	.889	1.125
Understanding_level	.109	.028	.136	3.851	.000	.906	1.104
HEC_Frequency_Class	.018	.023	.028	.769	.442	.855	1.170
a. Dependent Variable: Coexistence							

S3.5 Interpretation

The multicollinearity diagnostics indicated that the candidate predictors did not exhibit substantial multicollinearity. The observed VIF values were below the predefined threshold of 5, supporting the use of the predictor set in the subsequent binary logistic regression analysis.

S4. Binary Logistic Regression

S4.1 Objective

Binary logistic regression was performed to identify socio-demographic, spatial, conflict-related, and elephant-related factors associated with household willingness to coexist with elephants. The analysis estimated the direction and magnitude of association of each predictor with the probability of a positive coexistence response after accounting for the other predictors included in the model.

S4.2 Input Variables

The binary logistic regression used household-level willingness to coexist as the binary dependent variable. The dependent variable, Coexistence, was coded as 0 = No and 1 = Yes. Predictors screened for multicollinearity through the preliminary linear regression analysis were subsequently entered into the binary logistic regression model.

Dependent variable

Variable	SPSS name	Coding
Willingness to coexist	Coexistence	0 = No; 1 = Yes

Predictors

Predictor	SPSS variable
Village location	PA_NonPA_Village / Village_location
Age	Age_group_class
Tribal status	Tribal_NonTribal
Education	Education_Class
Occupation	Occupation_Class
Income	Income_Class
NTFP dependence	NTFP_Dependens

Distance from forest	Village_distance_From_Forest
Elephant group type	Elephant_Group_type
Understanding level	Understanding_level
HEC frequency	HEC_Frequency_Class

S4.3 Statistical Procedure

Binary logistic regression was conducted using the Enter method after assessment of multicollinearity. The model estimated the probability of a positive willingness-to-coexist response as a function of the selected predictors. Regression coefficients were estimated using maximum likelihood. Categorical predictors were represented using indicator variables with predefined reference categories.

The logistic model was expressed as:

$$\text{logit}(P)=\ln(1-PP)=\beta_0+\beta_1X_1+\beta_2X_2+\dots+\beta_nX_n$$

where represents the probability of a positive coexistence response, is the intercept, represents the regression coefficient for predictor, and represents the corresponding predictor.

S4.4 Model Evaluation

Model significance and goodness of fit were evaluated using the Omnibus Test of Model Coefficients, Hosmer–Lemeshow goodness-of-fit test, Cox and Snell and Nagelkerke pseudo- statistics, and the classification table. The discriminatory performance of the fitted model was additionally assessed using the area under the receiver operating characteristic curve (AUC).

S4.5 Regression Output

TABLE 2: TABLE OF BINARY LOGISTIC REGRESSION FOR COEXISTENCE BETA-VALUE

		Variables in the Equation					
		B	S.E.	Wald	Df	Sig.	Exp(B)
Step 1 ^a	PAVillages_NonPAVillages(1)	-1.713	.213	64.546	1	.000	.180
	Age_group_class			1.283	2	.526	
	Age_group_class(1)	.031	.222	.020	1	.889	1.032
	Age_group_class(2)	.219	.221	.983	1	.321	1.245
	Tribal_NonTribal(1)	-.620	.209	8.836	1	.003	.538
	Education_Class			2.985	2	.225	
	Education_Class(1)	-.285	.228	1.558	1	.212	.752
	Education_Class(2)	-.417	.244	2.933	1	.087	.659
	Occupation_Class			.483	2	.785	
	Occupation_Class(1)	.120	.245	.241	1	.624	1.128
	Occupation_Class(2)	.179	.317	.320	1	.572	1.196
	Income_Class(1)	-.338	.195	2.995	1	.084	.714
	NTPF_Dependent(1)	.092	.368	.063	1	.802	1.097
	Village_distance_Forest(1)	.570	.231	6.077	1	.014	1.769
	Elephant_Group_type(1)	-.895	.180	24.747	1	.000	.409
	Understanding_level			18.849	2	.000	
	Understanding_level(1)	.151	.362	.174	1	.677	1.163
	Understanding_level(2)	.918	.380	5.852	1	.016	2.505
	HEC_Frequency_Class			.821	2	.663	
	HEC_Frequency_Class(1)	.171	.236	.524	1	.469	1.187
	HEC_Frequency_Class(2)	.197	.229	.740	1	.390	1.218
	Constant	1.113	.564	3.896	1	.048	3.043
a. Variable(s) entered on step 1: PAVillages_NonPAVillages, Age_group_class, Tribal_NonTribal, Education_Class, Occupation_Class, Income_Class, NTPF_Dependent, Village_distance_Forest, Elephant_Group_type, Understanding_level, HEC_Frequency_Class.							
Reference categories are presented in the Methods section.							

S5. Probability of Coexistence Presence Index (PCPI) Calculation

S5.1 Objective

The Probability of Coexistence Presence Index (PCPI) was developed to translate the fitted binary logistic regression model into a quantitative measure of the predicted probability of positive coexistence with elephants. The PCPI was calculated using the regression coefficients obtained from the final binary logistic regression model and the corresponding coded predictor values for each surveyed household.

S5.2 Calculation of the Linear Predictor

For each household, the regression coefficients obtained from the binary logistic regression model were combined with the corresponding predictor values to calculate the linear predictor. The calculation incorporated the model intercept and the coefficients associated with village location, age, tribal status, education, occupation, income, NTFP dependence, distance from the forest, elephant group type, understanding level, and HEC frequency.

S5.3 Conversion to Predicted Probability

The linear predictor was transformed into a probability using the logistic transformation. The resulting probability represented the predicted likelihood of a positive coexistence response for each household.

$$\text{EXP}(X) \\ 1/(1+\text{EXP}(X))$$

S5.4 Household-level PCPI Calculation

Household-level PCPI values were calculated by applying the fitted logistic regression coefficients to the coded predictor values for each household. The resulting values ranged between 0 and 1 and represented the model-predicted probability of positive willingness to coexist.

S5.5 Aggregation to Village-level PCPI

Because the household-level survey data contained multiple observations for individual villages, the household-level PCPI estimates were aggregated by calculating the arithmetic mean of the predicted PCPI values for each village. The resulting village-level mean PCPI was used for spatial representation and subsequent coexistence-probability classification.

$$\text{PCPI}_{\text{village}} = n \sum_{i=1} \text{PCPI}_i / n$$

S5.6 Final Village-level Output

The final village-level PCPI dataset contained predicted coexistence probabilities for the surveyed villages. These values were subsequently linked to the village spatial layer for mapping and classification.

S6. Probability of Coexistence Index table village-wise Table

TABLE 3 PROBABILITY OF COEXISTENCE INDEX TABLE VILLAGE-WISE.

Village Name	Probability of Coexistence Presence Index (PCPI)
Badwahi	0.180645
Ghaghaud	0.181355
Bijhariya	0.185926
Pathari	0.189695
Kumhai	0.211562
Umariya	0.220067
Mala	0.220692
Sejwahi	0.224077
Karchchatola	0.228117

Kaseru	0.232625
Bara	0.243969
Majhouli	0.249281
Chansura	0.252894
Harrwah	0.261281
Khushmaha	0.265103
Pathor	0.266569
Guruwahi	0.268049
Majhkheta	0.275192
Bagdari	0.280704
Chhapraur	0.282328
Gohadi	0.285958
Hardi	0.287117
Saidol	0.297548
Bhadaura	0.298312
Chitraw	0.299984
Rakhiamodar	0.300595
Majhtola	0.307862
Majhouli	0.314742
Mardari	0.317757
Lurghuti-2	0.325469
Labahi	0.331176
Kharibadi, Bhamrahatola	0.336926
Badwahi	0.340771
Gaatta	0.354435
Sarsai	0.358933
Goli pahadi	0.360994
Kudari	0.363371
Basa	0.365521
Fulva	0.367774
Bagdara	0.369443
Lakhumar	0.375111
Mahaman	0.377036
Bagaiha	0.377754
Bamera	0.382842
Kothiya	0.384703
Kumharra	0.387935
Dukhariya	0.38832
Gadawah	0.391295
Ranchcha	0.393039
Sehra	0.411978
Kathli	0.425597
Medhaki	0.431925
Chitrauli	0.441109
Kundor	0.443125
Lakhnoti	0.462378

Purendol	0.463167
Dobha	0.468603
Madau	0.470828
Bagdo	0.474608
Tingi	0.478744
Harrai	0.491681
Jhalwar	0.495502
Chhapra Tola	0.504495
Patharwar	0.512791
Urdana	0.518504
Bastua	0.524443
Pondi	0.529735
Kathar	0.554803
Barhapan	0.556682
Bhathisarai	0.56423
Kuthaliya	0.570197
Gijohar	0.575664
Kota	0.576503
Dhodha	0.600315
Khodri-2	0.604821
Joba	0.608656
Laharpur	0.616678
Nimha	0.623356
Gobha	0.63472
Rasmohani	0.645054
Barutara	0.645382
Lukham Tola	0.646447
Aamgaon	0.650278
Vynkatnagar	0.651874
Nand Tola	0.654069
Malgo Tola	0.660772
Kukurgoda	0.661498
Beigavan	0.672871
Simaudh	0.678117
Amanar	0.681696
Bhamarkhoh	0.684293
Dhari	0.69077
Charkidol	0.692417
Rohilla Kachhar	0.699677
Mahua Tola	0.709862
Pataura Tola	0.71515
Pateriya Tola	0.7189
Khalbahara	0.721147
Kunda Tola	0.725766
Kudra Tola	0.728758
Mithul	0.730784

Sikta Tola	0.73107
Lahsuna	0.731401
Gobari	0.733624
Tanki	0.733681
Khohra	0.736158
Thehi	0.74539
Munda	0.746425
Antoli	0.748653
Banchachar	0.749834
Amha	0.754794
Khanda (Fulangi tola)	0.760237
Jeer	0.762798
Belgaon	0.763438
Amgawan (Dadra Tola)	0.763609
Khoh	0.771808
Tihki	0.774358
Chidmidi	0.778822
Hartala	0.780644
Jariyari	0.790345
Katurdona (mekal pahad)	0.792054
Pondi (Kapriya)	0.795472
Dhanera	0.809659
Bhelma, Mekal Pahad	0.811898
Cholna	0.815913
Ghori Ghat	0.82605
Kadamsara	0.833878
Sikarpur-Bagdumara	0.847432
Chhataini	0.849141
Bedra	0.849893
Bilai Kher	0.850652
Chatua (Bahratola)	0.853061
Kothiya	0.861465
Pipariya (Pondi)	0.864004
Jogibandh	0.864112
Takuhali	0.870642
Pathrehta	0.880071
Basahi	0.882517
Khursa	0.907411
Malauti	0.918255

